

- m On-demand bandwidth
- m Multimedia mail storage on the network itself
- m 2 Mbps broadband Internet access

Walking Tall—The Third Generation (3G)

CDMA (the technique, not the standard) is generally considered the future of cellular technology. CDMA based networks can carry a larger number of calls, are faster, more secure, and larger areas can be covered with fewer base stations. No wonder, then, that the two technologies that might dominate the 3G world are based on the CDMA principle.



A 3G Mobile Phone with streaming Video

Existing GSM networks will proceed to use a technique called W-CDMA (Wideband CDMA), which uses CDMA, but will allow for data transfer rates of about 2 Mbps if you sit in one place. In Japan, NTT DoCoMo rolled out their W-CDMA solution in 2001. They called it FOMA—*Freedom of Mobile Multimedia Access*. By 2004 it had covered 99 per cent of Japan.

Another technology based on W-CDMA is the *Universal Mobile Telephone System* (UMTS). It integrates with existing GSM infrastructures and provides data speeds of 1.99 Mbps. UMTS networks will use USIM (Universal SIM) cards, which are advanced versions of the regular SIM cards we use today. They are more